Reduced Description of Nonlinear Bend – Stretch Coupling in Carbon Dioxide Molecule

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Abstract

Strong resonance between bend and symmetric stretch carbon dioxide vibrations is used for separating the slow and fast processes. The slow process consists in the intermode energy redistribution. The one degree of freedom Hamiltonian is constructed for describing this process. Two stable and two unstable fixed points determine structure of the corresponding phase space. Periods and amplitudes of the energy redistribution is shown to be dependent on the initial intermode energy partitioning and phase difference (0 or π) of the bend and symmetric stretch vibrations. Chedin's potential is used for numerical calculations.

Key words: carbon dioxide, nonlinear intramolecular potential, intramolecular vibrational energy redistribution, coupled oscillators, Hamiltonian systems, phase space
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1 Introduction

Intramolecular nonlinear dynamics is important for various physical processes, e.g. chemical reactions, intramolecular energy redistribution, vibrational relaxation, etc. Many attempts were spent to understand general features of nonlinear intermolecular dynamics as well as the behavior of particular molecules [**?**]–[**?**].

Carbon dioxide plays an important role in environmental phenomena, industrial and scientific applications for laser systems and as a supercritical solvent

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in large scale technical extraction processes. Energy exchange is essential for most processes concerning laser operation and chemical reactions [1]–[4]. The energy paths of CO_2 lasers involve the energy transitions between all (symmetric and asymmetric stretch and bend) CO_2 modes as well as intermolecular energy exchange. Although energy transfer mechanisms were widely studied experimentally, theoretically many points remain unclear even for dilute systems. At high excitation, carbon dioxide vibrational dynamics is complicated by the nonlinearity of its intramolecular potential energy that is especially important due to strong resonance between the bend and symmetric stretch modes, the vibrational frequency of the latter being approximately twice as high as the former. Thus, it is important to investigate its intramolecular dynamics in detail.

The energy redistribution between the symmetric stretch and bend modes can strongly enhance carbon dioxide vibrational energy release to the solvent because there are many indications that the main pathways for vibrational energy relaxation lead via the lowest frequency vibrational modes [5]–[7]. At the same time, the equations describing the dynamics of CO_2 laser emission usually involve terms corresponding to energy flux directly from the symmetric stretch mode to the solvent without any connection to bend mode dynamics [8].

The paper is organized as follows. The next section is devoted to the description of the dynamical model and Chedin's potential [11]. In Sec. 3 the slow variables are identified. The dynamical equations for these variables provide evidence for energy redistribution between bend and symmetric stretch modes. The results of a numerical analysis of the slow variables dynamics and the comparison with numerical results that take into account fast motions are analyzed in Sec. 4.

2 Chedin's potential and dynamical model

CO_2 is a linear molecule with four internal degrees of freedom one of which is a quasirotation. Thus, its ground state potential energy can be represented in terms of three intramolecular coordinates. Chedin's coordinates R_1^*, R_2^* and R_3^* that correspond to the symmetric stretch, bend and asymmetric stretch deformations of the molecule, respectively, are determined [11] by the relations:

$$R_1^* = a_1 \left(2 - r_{\rm eq}\frac{r_{21}+r_{23}}{r_{21}r_{23}}\right), \quad R_2^* = a_2\alpha, \quad R_3^* = a_3 r_{\rm eq}\frac{r_{21}-r_{23}}{r_{21}r_{23}}, \qquad (1)$$

where r_{21} and r_{23} are the bond lengths between the oxygen (labeled 1 and 3) and carbon (labeled 2) atoms, $\pi - \alpha$ is the angle between these bonds, $r_{\rm eq}$

is the equilibrium bond length (the equilibrium distance between the carbon and oxygen atoms). Chedin's coordinates attain finite values at infinite bond lengths.

Chedin's potential can be represent as a sum of three terms

$$U = U_0 + U_{\rm sma} + U_{\rm cpl} \tag{2}$$

where

$$U_0 = F_1 R_1^{*2} + F_2 R_2^{*2} + F_3 R_3^{*2} \tag{3}$$

is the harmonic part of the potential and

$$\begin{aligned} U_{\rm sma} = {} & F_{300} R_1^{*3} + F_{400} R_1^{*4} + F_{500} R_1^{*5} + F_{600} R_1^{*6} + F_{040} R_2^{*4} + F_{060} R_2^{*6} \\ & + F_{004} R_3^{*4} + F_{006} R_3^{*6}, \end{aligned} \tag{4}$$

$$\begin{aligned} U_{\rm cpl} = {} & F_{120} R_1^{*} R_2^{*2} + F_{102} R_1^{*} R_3^{*2} + F_{220} R_1^{*2} R_2^{*2} + F_{202} R_1^{*2} R_3^{*2} \\ & + F_{022} R_2^{*2} R_3^{*2} + F_{320} R_1^{*3} R_2^{*2} + F_{302} R_1^{*3} R_3^{*2} + F_{140} R_1^{*} R_2^{*4} \\ & + F_{104} R_1^{*} R_3^{*4} + F_{420} R_1^{*4} R_2^{*2} + F_{402} R_1^{*4} R_3^{*2} + F_{240} R_1^{*2} R_2^{*4} \\ & + F_{204} R_1^{*2} R_3^{*4} + F_{042} R_2^{*4} R_3^{*2} + F_{024} R_2^{*2} R_3^{*4} \\ & + F_{122} R_1^{*} R_2^{*2} R_3^{*2} + F_{222} R_1^{*2} R_2^{*2} R_3^{*2} \end{aligned} \tag{5}$$

stand for the anharmonic single mode and intermode coupling parts of the potential, correspondingly. Constants that appear in Chedin's potential are determined from spectroscopic measurements [11].

The kinetic energy of the system under consideration can be written as follows:

$$K = \frac{m_{\rm O}}{4} (\dot{q}_1^2 + \dot{q}_2^2 + \dot{q}_3^2 + q_2^2 \dot{\varphi}^2), \tag{6}$$

where q_1 and q_3 are the normal coordinates for symmetric and asymmetric stretches, respectively, q_2 and φ are the coordinates for doubly degenerate bend. These coordinates are expressed in terms of coordinates $\{x_i,\ y_i,\ z_i\}$ $(i = 1, 2, 3)$ of CO_2 atoms in the Eckart frame [12,13] by the following formulae

$$q_1 = x_3 - x_1 - 2r_{\rm eq}, \quad q_3 = \gamma(x_3 + x_1),$$

$$q_2 = 2\gamma\sqrt{y_3^2 + z_3^2}, \quad \varphi = \arctan y_3/z_3 \tag{7}$$

$$\gamma = \sqrt{1 + 2m_{\rm O}/m_{\rm C}}, \tag{8}$$

and $m_{\rm C}$ and $m_{\rm O}$ are the masses of carbon and oxygen atoms, respectively.

To write down the equations of motion it is desirable to express the coordinates (7) in terms of Chedin's coordinates (1). Accomplishing this we take into consideration the relations

$$r_{21}^2 = \frac{1}{4}\left[(q_1 + 2r_{\rm eq} - \gamma q_3)^2 + \gamma^2 q_2^2\right],$$
$$r_{23}^2 = \frac{1}{4}\left[(q_1 + 2r_{\rm eq} + \gamma q_3)^2 + \gamma^2 q_2^2\right]. \qquad (9)$$

It is worth to investigate nonlinear effects caused exclusively by the Chedin's potential nonlinearity. In this case the kinematic nonlinearities contained in Eqs. (9) can be neglected and the expressions for the distances r_{21} and r_{23} assume the linear form

$$r_{21} = r_{\rm eq} + (q_1 - \gamma q_3)/2,$$
$$r_{23} = r_{\rm eq} + (q_1 + \gamma q_3)/2. \qquad (10)$$

For the following consideration it is convenient to introduce new coordinates

$$R_1 = a_1^{-1} R_1^*, \quad R_2 = a_2^{-1} R_2^*, \quad R_3 = a_3^{-1} R_3^*, \qquad (11)$$

to which we also refer as Chedin's ones. In the linear approximation

$$R_1 = -\frac{q_1}{r_{\rm eq}}, \quad R_2 = \gamma\frac{q_2}{r_{\rm eq}}, \quad R_3 = -\gamma\frac{q_3}{r_{\rm eq}} \qquad (12)$$

and the kinetic energy of the system is represented in terms the time derivatives of Chedin's coordinates

$$K = \frac{1}{2}(\mu_1 \dot{R}_1^2 + \mu_2 \dot{R}_2^2 + \mu_3 \dot{R}_3^2 + \mu_1 R_1^2 \dot{\varphi}^2), \qquad (13)$$

where inertia coefficients

$$\mu_1 = \frac{m_O r_{\rm eq}^2}{2}, \quad \mu_2 = \mu_3 = \frac{m_O r_{\rm eq}^2}{2\gamma^2}. \qquad (14)$$

The carbon dioxide model as defined by Eqs. (1) through (5) and (13) takes into account the *linear* kinematic interaction between stretch and bend *local* modes that was earlier investigated for water- and benzene-like molecules (see Refs. [14]-[16] and references therein). The corresponding terms can be singled out from the kinetic energy Eq. (13) after the transition to local coordinates.

These terms give *linear* contributions to the equations of motion and lead to beats in the *local* coordinates time evolution. Of course, no such beats appear in the dynamics of normal coordinates if nonlinear interactions are not taken into account.

The equations of motion contain kinematic nonlinear terms originating from the kinetic energy as well. This nonlinearity is shown to be important for the stretch – bend coupling, e. g. for deuterated water molecules [17]. However, its consideration is beyond the scope of the current paper. Furthermore, only in-plane motion will be considered here. It means we neglect the quasirotation (the last term in Eq. (13) for kinetic energy).

The expressions for the kinetic (13) and potential (2) energies are used to construct the equations of motion for CO_2 in terms of Chedin's coordinates (subscript j runs from 1 through 3):

$$\ddot{R}_j + \omega_j^2 R_j = \omega_j^2 f_j, \tag{15}$$

where

$$\omega_j = 2\pi\nu_j = a_j\sqrt{2F_j/\mu_j} \tag{16}$$

correspond to the harmonic frequencies of symmetric stretch ($\nu_1 = 1352\text{cm}^{-1}$), bend ($\nu_2 = 672\text{cm}^{-1}$) and asymmetric stretch ($\nu_3 = 2393\text{cm}^{-1}$), and

$$f_j = -\frac{1}{2F_j a_j^2}\frac{\partial}{\partial R_j}(U_{\text{sma}} + U_{\text{cpl}}) \tag{17}$$

are dimensionless forces. In accordance with Eq. (2)–(5) these forces are functions of Chedin's coordinates.

3 Dynamics of slow variables

3.1 Carbon dioxide dynamics in amplitude and phase variables

For not too strong excitations ($R_j \sim 0.1$) we can consider the largest contributions only in the equations of motion (15). Then the potential energy and the dimensionless forces read

$$U = U_0 + F_{300}a_1^3R_1^3 + F_{120}a_1a_2^2R_1R_2^2 + F_{102}a_1a_3^2R_1R_3^2, \tag{18}$$

$$f_1 = k_1'R_1^2 + k_1R_2^2 + k_1''R_3^2, \quad f_2 = k_2R_1R_2, \quad f_3 = k_3R_1R_3, \tag{19}$$

where coefficients k_j are given in Table 1.

Therefore, Eqs. (15) take the form

$$\begin{aligned}
&\ddot{R}_1 + \omega_1^2 R_1 = \omega_1^2 (k_1' R_1^2 + k_1 R_2^2 + k_1'' R_3^2), \\
&\ddot{R}_2 + \omega_2^2 R_2 = \omega_2^2 k_2 R_1 R_2, \\
&\ddot{R}_3 + \omega_3^2 R_3 = \omega_3^2 k_3 R_1 R_3.
\end{aligned} \tag{20}$$

It is convenient to introduce complex variables [18] $c_j = R_j + i\omega_j^{-1}\dot{R}_j = A_j \exp[-i\varphi_j]$ and rewrite Eqs. (20) in terms of amplitudes A_j and phases φ_j

$$\begin{aligned}
\dot{A}_1 &= -\omega_1 \sin\varphi_1 (k_1 A_2^2 \cos^2\varphi_2 + k_1' A_1^2 \cos^2\varphi_1 + k_1'' A_3^2 \cos^2\varphi_3), \\
\dot{\varphi}_1 &= \omega_1 - \frac{\omega_1}{A_1} \cos\varphi_1 (k_1 A_2^2 \cos^2\varphi_2 + k_1' A_1^2 \cos^2\varphi_1 + k_1'' A_3^2 \cos^2\varphi_3), \\
\dot{A}_2 &= -\frac{\omega_2 k_2}{2} A_1 A_2 \cos\varphi_1 \sin^2\varphi_2, \\
\dot{\varphi}_2 &= \omega_2 - \omega_2 k_2 A_1 \cos\varphi_1 \cos^2\varphi_2, \\
\dot{A}_3 &= -\frac{\omega_3 k_3}{2} A_1 A_3 \cos\varphi_1 \sin^2\varphi_3, \\
\dot{\varphi}_3 &= \omega_3 - \omega_3 k_3 A_1 \cos\varphi_1 \cos^2\varphi_3.
\end{aligned} \tag{21}$$

The harmonic values of mode energies are proportional to squared amplitudes

$$E_j = \frac{\mu_j}{2}\dot{R}_j^2 + F_j a_j^2 R_j^2 = \frac{\mu_j}{2}(\dot{R}_j^2 + \omega_j^2 R_j^2) = \frac{\mu_j \omega_j^2}{2} c_j c_j^* = \frac{\mu_j \omega_j^2}{2} A_j^2, \tag{22}$$

where Eq. (16) is used.

As the ratio of the harmonic frequencies of symmetric stretch and bend vibrations is approximately equal to 2, the strong resonance between these vibrations allows one to separate the fast and slow terms in the equations of motion. The slow terms contain the phase difference

$$\Phi = \varphi_1 - 2\varphi_2. \tag{23}$$

Thus, it is possible to average Eqs. (21) over the time segments of the order of the bend vibrational period $2\pi/\omega_2$ to remove all the terms that contain fast phases φ_1, $3\varphi_1$, $(\varphi_1 + 2\varphi_2)$ and $(\varphi_1 \pm 2\varphi_3)$. Then Eqs. (21) take the form

$$\dot{A}_1 = -\frac{\omega_1 k_1}{4} A_2^2 \sin\Phi, \qquad \dot{\varphi}_1 = \omega_1 - \frac{\omega_1 k_1}{4A_1} A_2^2 \cos\Phi, \tag{24}$$

$$\dot{A}_2 = \frac{\omega_2 k_2}{4} A_1 A_2 \sin\Phi, \qquad \dot{\varphi}_2 = \omega_2 - \frac{\omega_2 k_2}{4} A_1 \cos\Phi, \tag{25}$$

$$\dot{A}_3 = 0, \qquad \dot{\varphi}_3 = \omega_3. \tag{26}$$

For smoothed amplitudes and phases we preserve the same notation as in Eqs. (21).

Eqs. (26) show that in this approximation the asymmetric mode is harmonic and independent from the others. The problem is reduced to the treatment of coupling solely between the symmetric stretch and bend modes.

The equations for time derivatives of amplitudes A_1 and A_2 after eliminating $\sin\Phi$ can be integrated to obtain the relationship between the smoothed amplitudes

$$A_1^2 + kA_2^2 = e_0, \quad k = \frac{\omega_1 k_1}{\omega_2 k_2} \tag{27}$$

with the integration constant

$$e_0 = A_{10}^2 + kA_{20}^2 = e_{10} + \frac{\omega_1}{2\omega_2} e_{20} \tag{28}$$

represented through the initial values of the amplitudes A_{10} and A_{20} or the initial mode harmonic energies E_{10} and E_{20} in the dimensionless form

$$e_{10} = 2E_{10}/\mu_1\omega_1^2, \quad e_{20} = 2E_{20}/\mu_1\omega_1^2. \tag{29}$$

The relationship

$$2\mu_1\omega_1^2 k_1 = \mu_2\omega_2^2 k_2 \tag{30}$$

was taken into account for deriving Eq. (28). $k \approx 0.0678$ for the model under consideration.

Thus, energies are determined in units of $\mu_1\omega_1^2/2 = a_1^2 F_1$ (the potential constant $F_1 = 710\text{cm}^{-1} = 8.49$ kJ/mol) and for arbitrary time

$$e_1 = A_1^2, \quad e_2 = k(2\omega_2/\omega_1)A_2^2. \tag{31}$$

Indeed, Eq. (27) reflects the energy conservation in terms of the smoothed variables because the coefficient in front of e_{20} in Eq. (28) is close to 1 (it is approximately equal to 1.006). This coefficient is determined by harmonic characteristics of the system and becomes equal to one for the ideal resonance

condition $\omega_1 = 2\omega_2$ although the energy excess ($e_0 > e_{10} + e_{20}$) is due to the nonlinear terms.

Eq. (27) can be used for simplifying the description to the level of a single degree of freedom

$$\dot{A}_1 = -\frac{\omega_2 k_2}{4}(e_0 - A_1^2)\sin\Phi,$$
$$\dot{\Phi} = \Omega - \frac{\omega_2 k_2}{4A_1}(e_0 - 3A_1^2)\cos\Phi, \qquad \Omega = \omega_1 - 2\omega_2. \tag{32}$$

As Eqs. (32) are strongly nonlinear, the behavior of the amplitude A_1 and the phase difference Φ obviously depend on the initial conditions: the initial energy, its partitioning among the modes (A_{10}/A_{20}), and the initial phase difference (Φ_0). After Eqs. (32) are integrated the time dependence of the bending mode amplitude A_2 and the smoothed phases φ_1 and φ_2 can be determined by Eqs. (24), (25) and (27).

3.2 Hamiltonian description of slow variables

Eqs. (32) can be considered as the amplitude-phase representation of equations of motion of an artificial one degree of freedom system with Hamiltonian

$$H = \frac{\Omega}{2}[(x^2 + y^2) + bx(x^2 + y^2 - e_0)], \qquad b = \frac{\omega_2 k_2}{2\Omega}. \tag{33}$$

Actually, the canonical equations corresponding to this Hamiltonian

$$\dot{x} = \Omega(1 + bx)y, \qquad \dot{y} = -\Omega[(1 + bx)x + (b/2)(x^2 + y^2 - e_0) \tag{34}$$

coincide with Eqs. (32) after the transition to the amplitude-phase representation $x + iy = A_1 \exp[-i\Phi]$. Thus, $x = A_1\cos\Phi$ and $y = -A_1\sin\Phi$ are canonically conjugated variables.

The integral curves $H = Ce_0\Omega/2$ ($C = const$) in the phase space of the system are given by the equation

$$x^2 + y^2 = \frac{C + bx}{\Omega + 2bx}e_0 \tag{35}$$

The set of these curves for different values of C is shown in fig. 1. *What are the conditions for Fig. 1?*

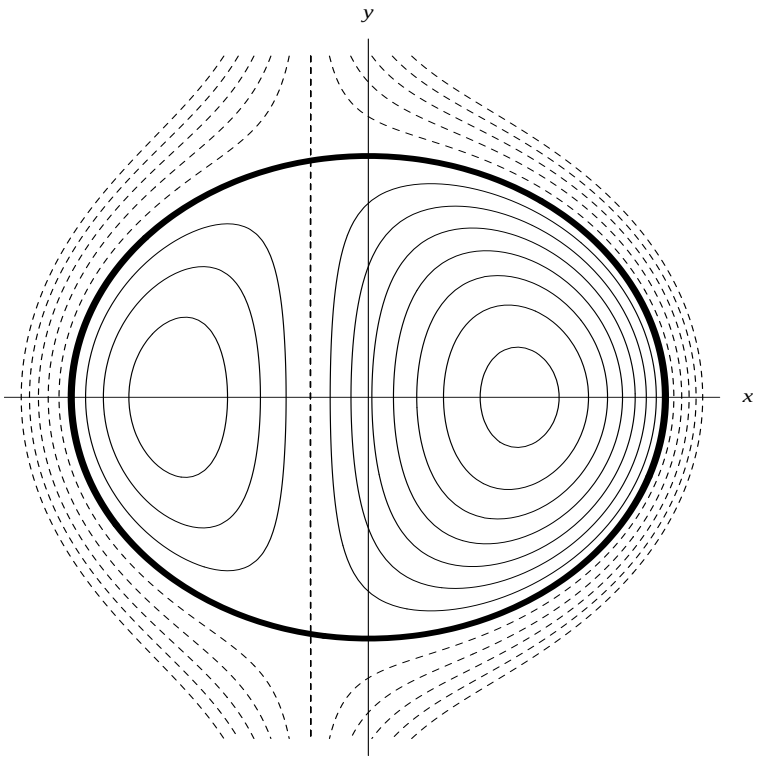


Fig. 1. Integration curves

At not very low excitation ($e_0 > b^{-1}$) Eqs. (34) have 4 fixed points $(\bar{x}, \bar{y})$:

$$\bar{x}_{1,2} = -(3b)^{-1}\left(1 \pm \sqrt{1 + 3b^2 e_0}\right), \quad \bar{y}_{1,2} = 0; \tag{36}$$

$$\bar{x}_{3,4} = -b^{-1}, \quad \bar{y}_{3,4} = \mp\sqrt{e_0 - b^{-2}}. \tag{37}$$

As for the model under consideration $b = 74.3 >> 1$, at $e_0 \geq 1$ with a high accuracy

$$\bar{x}_{1,2} = \mp\sqrt{e_0/3} - 1/(3b).$$

Fixed points 1 and 2 are stable points while 3 and 4 are unstable ones. The system phase space is divided by separatrixes (thick lines in Fig. 1) into four regions. Both the circle $x^2 + y^2 = e_0$ and vertical $x = -b^{-1}$ separatrixis correspond to $C = 1$.

In accordance with Eq. (27)

$$x^2 + y^2 = e_0 - kA_2^2 \leq e_0. \tag{38}$$

It means that the regions outside the circle separatrix are forbidden for the original dynamical model.

The region $x > -b^{-1}$ inside the circle separatrix contains two types of integral curves depending on whether it surrounds the coordinate system origin ($x = 0, y = 0$) or not. The phase difference Φ is a periodic function of time for the latter and monotonically increasing one for the former. In the region $x < -b^{-1}$ the oscillating phase difference behavior is only observed.

At a given value of total excitation energy e_0 the integration constant C selects the phase trajectory with initial energy $e_{10} = x_0^2$ at $\Phi_0 = 0$ or π ($y_0 = 0$)

$$C = (x_0^2/e_0)(1 + bx_0) - bx_0. \tag{39}$$

It is enough to consider the segment $\bar{x}_1 < x_0 < \bar{x}_2$ for creating all the phase trajectories inside the circle separatrix. Each of them intersects x-axis at the second point

$$x_1 = \frac{1 + bx_0}{2b} \pm \frac{1}{2b}\sqrt{(1 - bx_0)^2 - 4b^2x_0^2 + e_0}, \tag{40}$$

where the upper sign must be taken for $-b^{-1} < x_0 < \bar{x}_2$. The difference $|x_1^2 - x_0^2|/2$ determines the amplitude of the energy exchange that is shown in Fig. 3 as a function of x_0 for several values of e_0. The strongest energy exchange takes place for trajectories close to the separatrixes.

Considering the equations of motion in the vicinity of the stable points it is easy to obtain the minimal values of periods of the intermode energy exchange

$$\bar{T}_{1,2} = \frac{2\pi}{\Omega|1 + b\bar{x}_{1,2}|} = \frac{2\pi}{\Omega[\sqrt{1 + 3b^2e_0} \pm (2/3)]} \simeq \frac{2\pi}{k_2\omega_2\sqrt{3e_0}} = \frac{2T_2}{k_2\sqrt{3e_0}}, \tag{41}$$

where $T_2 = 2\pi/\omega_2$ is a bend vibrational period.

I've encountered a difficulty here because the estimation of the period $\bar{T}$ is not correct, too short if I take the value of k_2 in accordance with our article submitted to NPCS, Eqs.(32). Eqs. (27) and (30) in the current paper are not compatible with the k_j-values as well, and with k=0.4754. Probably, you'l fix this point more easy than me.

Two constants, e_0 and b, determine the structure of the phase space. At very low excitation energies $e_0 < b^{-1}$ only the vertical separatrix and two first fixed points survive (see Fig. 2). The states of the original system are disposed inside of the circle $x^2 + y^2 = e_0$, which here is not a separatrix from topological point of view. We do not consider this case in more detail because low excitation energies are of no importance for practical applications.

Additional insight in the process of intermolecular energy redistribution can be achieved if we multiply first and second of Eqs. (20) with $\mu_1\dot{R}_1$ and $\mu_2\dot{R}_2$, respectively:

$$\begin{aligned}\dot{E}_1 &= \mu_1\omega_1^2 k_1 R_2^2 \dot{R}_1,\\ \dot{E}_2 &= 2\mu_2\omega_2^2 k_2 R_1 R_2 \dot{R}_2.\end{aligned} \tag{42}$$

After the transition to the smoothed amplitude and phase variables and neglecting by $\dot{A}_j << \omega_j A_j$ we get in a dimensionless form

$$\begin{aligned}\dot{e}_1 &= \frac{k_1}{2}\omega_1 A_2^2 A_1 \sin\Phi,\\ \dot{e}_2 &= -k_1\omega_2 A_1 A_2^2 \sin\Phi.\end{aligned} \tag{43}$$

The mode energy derivatives are of opposite sign and proportional to the symmetric stretch amplitude and to the square of the bending mode amplitude. Thus, the intermode energy exchange becomes very slow when the bending mode amplitude attains small values. The phase difference controls the direction and intensity of the intramolecular energy flow.

It is desirable to have illustration figure here, something like Fig. 3 from NPCS.

4 Conclusions

The contribution of the nonlinear terms of Chedin's potential to the dynamics of a carbon dioxide molecule is investigated. Due to a strong resonance between the bend and symmetric stretch mode efficient energy exchange between these modes exists and can be described in terms of slow variables, i. e. the amplitude (or energy) of the symmetric stretch (or, alternatively, bending) mode and the phase difference between them. It is shown that the slow variables dynamics is well separated from the asymmetric stretch mode.

After transition to slow variables the equations of motion of the bend and symmetric stretch modes are reduced to the form of a single degree of freedom that allows us to analyze the carbon dioxide classical dynamics in a great detail. The comparison with solutions of the full dynamical equations indicates that the slow variables equations well represent the main features of carbon dioxide vibrational dynamics.

The diversity of slow variables dynamics consists in wide variations in amplitude and period of intermode energy exchange that is completely determined

by initial conditions, which include the phase difference and the partitioning of energy among modes. The only term of the potential energy that is linear in the symmetric stretch and quadratic in the bend mode coordinates is responsible for significant low frequency intermode energy exchange. The other potential energy terms superimpose high frequency variations on this main behavior.

The energy exchange spans from complete energy transfer back and forth between the symmetric stretch and bend to the absence of such an exchange depending on the initial conditions.

All these findings are important for understanding the process of carbon dioxide internal energy exchange with bath molecules. It is a conventional point of view that the major part of the vibrational energy flux to the translational and rotational degrees of freedom of surrounding molecules occurs via low frequency modes, i. e. the bending mode in the case of carbon dioxide. The resonance conditions enhance the deactivation of the symmetric stretch via the energy exchange with the bend.

Nevertheless, at a variety of initial conditions the symmetric stretch – bend energy exchange is not efficient. Additionally, the bend excitation lasts only a small portion of the energy exchange period (see Figs. 2 and 3). Thus, it is of great importance to study the influence of the interaction with bath molecules on the process of energy redistribution among internal states of carbon dioxide molecules.

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